

Introduction to Blockchain and Smart Contracts

Lecture 13: Exchanges, CEX Proofs

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Crypto Exchanges

- Important functions:
 - **Price discovery:** rate at which one currency can be exchanged for another
 - **Matching:** Connecting buyers and sellers
 - **Custody/Settlement**
 - **Liquidity provisions**

Crypto Exchanges

- Types of exchanges —
 - **Decentralised:** Smart contract based trading system. Wallets are self-custodial.
 - Eg: Uniswap
 - Two broad categories:
 - Orders can be posted either off-chain or on-chain using **order books** but settlement is always on-chain
 - Users trade against **liquidity pools** rather than traditional order books (**A**utomated **M**arket **M**aker)
 - No counterparty; price rate algorithmically fixed; Liquidity providers — anyone can put their money into the pool to earn a share of the trading fee.

Decentralised Exchange

- Example: Anita wants to swap ETH for USDC on a DEX
 - Anita opens a wallet such as **MetaMask**.
 - She connects it to a DEX interface.
 - She selects “swap 1 ETH for USDC.”
 - The DEX contract quotes an output based on current pool state.
 - Anita signs a transaction with her wallet.
 - The blockchain executes the swap.
 - Anita receives USDC directly into her wallet.

Crypto Exchanges: Centralised

- Types of exchanges —
 - **Centralised:** Binance, Coinbase, ...
 - Wallets under exchange control, internal ledger, user accounts, ..
 - Custodian of wallets and deposited coins
 - Easier to regulate

Proof Requirements for CEXs

- Several security scares:
 - Bank run: When simultaneously many people want to withdraw
 - Bank owners are crooks
 - Insider attack ..
- Proofs of
 - Assets, Reserves, Liabilities, Solvency

Proof of Asset

- **The exchange proves it controls certain assets. It may:**
 - Publish wallet address
 - Show balances on-chain
 - **Sign a challenge message proving it controls those addresses**
 - Digitally sign a message; if the signature verifies then CEX owns the address
- What it doesn't prove is:
 - These are comprehensive, unencumbered, no hidden debts, liabilities are smaller than held assets

Proof of Reserves

- Broader than proof of assets
- Q: Does CEX hold enough reserves to cover what it owes to users?
- Contains three parts:
 - **Reserves:** What assets exchange controls on-chain (Proof of Assets)
 - **Liabilities:** What the exchange owes to customers (Proof of Liabilities)
 - Need to cryptographically commit the proof of liabilities
 - Reserves $\geq$ Customer Liabilities

Proof of Liabilities

- Take a snapshot of all customer balances
- Cryptographically commit by creating a Merkle tree of customer balances and publishing the root hash.
- Each customer can independently verify their inclusion
- But, can user convince themselves beyond their inclusion that fractional reserve ≥ 1 ?
 - Change Merke tree to include hash and **sum of balances in that subtree**

Oracles

- Sources of external (off-the-chain) information
- Push vs Pull Oracles
 - Push: Periodically publish the data on-chain
 - Pull: Responds only when a contract/user requests for data
- Oracles — collude, no data provenance,
 - **Solution** — source diversity, decentralised, time-weighted, staked collaterals, threshold signatures (multiple oracles must sign)

Sidechains, Statechains, L2 Rollups

Ethereum is SLOW

- **Sidechain:** Move Txns to separate chain and occasionally connect back to Ethereum
 - Have their own validators and security/consensus measures
 - It has its own block production rules and its own bridge to move assets in or out
- **State channels & Rollups:** Keep the activity off-chain between a small set of participants and settlements on Ethereum

Sidechains

- How do they work? Examples include: Polygon PoS [Used by Nike, Starbucks, ..]
 - Anita locks 10 ETH in a **bridge contract** on Ethereum
 - The sidechain bridge mints or releases 10 “**wrapped ETH**” for Anita on the side chain
 - Anita now transacts on the sidechain with cheaper costs
 - She burns/releases the wrapped asset on the side chain
 - The bridge unlocks the original ETH on Ethereum
 - Of course accounting for Bridge fees, de-pegging of wrapped assets, transaction fee on the main-net, etc.

Polygon PoS — Sidechain

- Two layered architecture:
 - Bor layer: The execution/block production layer
 - Heimdall layer: The consensus/validator layer; uses CometBFT
- The Ethereum layer is used to host staking contracts where Polygon checkpoints are posted